

2026 W23 Report

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Paper Title	Problems identified	Innovative points	Solutions	Effects	Project Transferability
Physics-Informed Deep Learning for Musculoskeletal Modeling: Predicting Muscle Forces and Joint Kinematics From Surface EMG[1]	Pure data-driven deep learning models function as 'black boxes' lacking physical interpretability and biomechanical compliance. Conversely, traditional physics-based models exhibit high computational latency, rendering them unsuitable for real-time applications.	The proposition of a physics-informed deep learning (PIDL) framework. It embeds macroscopic physical principles (equations of motion) as soft constraints directly into the loss function of the neural network.	Utilising a lightweight convolutional neural network (CNN) where the loss function combines standard data regression errors with a physics-based penalty. This penalty evaluates whether the predicted muscle forces and joint kinematics conform to classical rigid body dynamics.	The model demonstrated superior prediction accuracy compared to pure data-driven baselines. It also exhibited accelerated convergence, reduced dependency on large training datasets, and robust cross-subject generalisation.	Although hardware limitations restrict continuous tracking, the core concept of 'physics-based loss penalties' can be adapted. Applying biomechanical constraints during model training can act as an algorithmic filter, mitigating severe hardware noise and stabilising discrete gesture classification across varied user populations.
TraHGR: Transformer for Hand Gesture Recognition via Electromyography[2]	Classifying hand movements based on sparse multichannel sEMG signals is challenging because existing deep learning approaches typically utilize only a single model, which struggles to sufficiently extract representative spatio-temporal features.	The proposition of a hybrid framework based on the Transformer architecture (a model reliant on self-attention mechanisms) to simultaneously capture both temporal and spatio-temporal representations from sEMG signals.	Developing the TraHGR architecture featuring two parallel processing paths: a Temporal Transformer Network (TNet) handling temporal patches, and a Feature Transformer Network (FNet) handling featural patches. The outputs are subsequently integrated via a linear fusion layer.	The TraHGR-Huge variant achieved a high recognition accuracy of 86.18% on the Ninapro DB2 dataset (49 gestures, 200ms window), significantly outperforming state-of-the-art CNNs, RNNs, and traditional machine learning classifiers.	The massive parameter count (>846,000) of the Transformer model mathematically prohibits direct deployment on low-cost microcontrollers.

Continuous Motion Intention Prediction Using sEMG for Upper-Limb Rehabilitation: A Systematic Review of Model-Based and Model-Free Approaches[3]	Existing continuous motion intention prediction algorithms struggle with real-world robustness (e.g., electrode shift, sweating), poor cross-subject generalization, and the inherent trade-off between Model-Based (MB - computationally heavy) and Model-Free (MF - black-box, prone to overfitting) paradigms.	Systematically synthesizing 186 studies to propose that the future trajectory relies on combining the physiological interpretability of MB approaches with the advanced feature extraction of MF approaches, alongside leveraging Muscle Synergy (MS) and Motor Unit (MU) features.	Categorizing a decade of research by joint type and algorithmic methodology, and recommending the integration of closed-loop feedback, attention mechanisms, and transfer learning to mitigate cumulative errors and inter-subject variability.	Concludes that multi-sensor fusion, High-Density sEMG (HD-sEMG), and hybrid MB-MF models significantly enhance prediction accuracy and robustness compared to standalone traditional machine learning or rigid musculoskeletal models.	High-Density sEMG is physically incompatible with our dual-channel setup. However, the core Muscle Synergy (MS) principle is transferable. Computing cross-channel correlations (e.g., amplitude ratios) filters mechanical sliding noise. Furthermore, the confirmed difficulty of cross-subject generalisation justifies using the visual 'Automated Labelling Oracle' to bypass expensive transfer learning via rapid, automated individual recalibration.
Real-Time Hand Gesture Recognition by Decoding Motor Unit Discharges Across Multiple Motor Tasks From Surface Electromyography[4]	Existing surface electromyography (sEMG) decomposition techniques are primarily restricted to offline analysis or isometric contractions, lacking the capacity to decode motor unit (MU) activities across multiple dynamic hand gestures in real-time.	The formulation of a motion-wise motor unit decoding framework capable of tracing MU discharges across multiple (>10) motor tasks in real-time, effectively transitioning MU-based control from static to dynamic operational environments.	The methodology segments high-density sEMG (HD-sEMG) signals by specific motion phases. It applies a Convolution Kernel Compensation (CKC) algorithm to compute local MU filters for each segment, which are subsequently reused for global real-time decomposition to extract neural drive features.	Evaluated on 128-channel HD-sEMG arrays across 12 gestures, the real-time MU decomposition framework achieved classification accuracies that significantly outperformed standard macroscopic time-domain features, demonstrating high algorithmic robustness.	We cannot apply this motor unit decoding because it relies on 128 channels, whilst we only have two. Since our setup can only capture the overall, noisy muscle signal, advanced signal separation is a dead end.

The Fifth Main Dynamic Factor: Skin Temperature and Its Effects on EMG-Based Gesture Recognition (EMBC 2025)[5]	Existing surface electromyography (sEMG) gesture recognition models frequently fail in real-world scenarios due to dynamic factors. Whilst factors like electrode shift and muscle fatigue are well-documented, the impact of skin temperature is systematically overlooked.	The explicit identification and quantification of skin temperature as the 'fifth dynamic factor', demonstrating its direct interference with the physiological properties of the sEMG signal.	Recording synchronised sEMG data from 12 participants across three controlled temperature states and evaluating how these thermal variations alter signal features and classifier accuracy.	Cold conditions objectively increased signal amplitude whilst decreasing median frequency. Models trained solely on baseline data experienced a severe accuracy drop (from 84.76% to 66.86%) when tested in cold conditions.	Confirmation of dynamic physical limits that dry electrodes lack conductive gel, making their impedance highly sensitive to skin moisture and temperature. This paper proves that a static, pre-trained model will inevitably fail as the user's skin temperature changes. It further validates the absolute necessity of the visual 'Automated Labelling Oracle' for rapid, on-the-fly recalibration under current physical conditions.
Decoding sEMG Under Non-Ideal Conditions Toward Robust Muscle-Machine Interface Control (IROS 2023)[6]	Evaluating algorithmic strategies under pristine laboratory conditions fails to guarantee robustness in practical robotic rehabilitation, where systems confront multiple, simultaneous non-ideal conditions (e.g., electrode shifts, posture changes, fatigue).	A systematic, multi-variable evaluation framework applying an Extreme Learning Machine (ELM) methodology specifically tested against five distinct, real-world non-ideal physical interferences.	Extracting standard time and frequency domain features and processing them through an ELM classifier to recognise seven hand gestures, deliberately measuring accuracy degradation across varied arm postures, fatigue levels, and cross-day sessions.	The ELM methodology maintained robust classification under specific isolated non-ideal conditions compared to traditional algorithms, though inter-subject and inter-day variabilities remained significant algorithmic bottlenecks.	The findings demonstrate that complex deep learning is not mandatory for robust control. Standard time-domain features combined with a lightweight classifier (like ELM or SVM) can withstand moderate physical interference. This supports the strategy of deploying a lightweight traditional classifier on the microcontroller.

Comparison of Wrist and Forearm EMG for Multi-day Biometric Authentication (EMBC 2023)[7]	Current sEMG-based biometric authentication predominantly utilises forearm data, inherited from prosthetic research. However, the robustness and signal stability of wrist-worn configurations across multiple days remains an unresolved physical problem.	A direct, objective comparative analysis of spatial sensor placement (wrist versus proximal forearm) regarding long-term signal stability and multi-day classification degradation.	Utilising the open GRABMyo dataset (43 participants) to extract features and evaluate cross-day classification performance, specifically separating data collected on days 1, 8, and 29 to test reliability.	Forearm sEMG signals objectively provided superior stability and significantly higher multi-day accuracy compared to wrist sEMG, which exhibited severe degradation due to tendon-dominated anatomy at the wrist.	Although focused on biometrics, the underlying physical law applies to gesture control. The wrist lacks sufficient muscle belly volume, rendering signals weak and highly susceptible to posture shifts. Our dual-channel dry electrodes must be physically deployed on the proximal forearm to ensure basic signal viability.
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sEMG related GitHub Repositories	System Architecture	Hardware	Software	Demo/Prototype
facebookresearch / generic-neuromotor-interface[8]	A framework that decodes sEMG signals into continuous hand movements and discrete gestures.	High-density sEMG (HD-sEMG) arrays.	Python, PyTorch.	Scripts for offline testing to predict handwriting and wrist actions.
LibEMG / libemg[9]	A modular library for real-time myoelectric control, handling data streaming, filtering, feature extraction, and classification.	Hardware-agnostic (compatible with custom boards like ESP32 via UDP/Serial, or commercial armbands).	Python, scikit-learn, NumPy.	A real-time online user interface for testing gesture recognition.
tsagkas / sEMG-HandGestureRecognition[10]	A deep learning pipeline that uses Convolutional Neural Networks (CNNs) to classify hand gestures from raw sEMG data.	Standard sEMG sensors (e.g., Myo armband) based on public datasets.	Python, TensorFlow / Keras.	Scripts for training the CNN model and evaluating gesture classification accuracy.
josecyc / Neurotron[11]	A cross-modal system using Long Short-Term Memory (LSTM) networks to predict 3D hand poses by aligning sEMG signals with optical depth labels.	Myo armband (8-channel sEMG) and Leap Motion controller (spatial label oracle).	Python, TensorFlow / Keras.	A VR/AR interface predicting 21 hand keypoints driven solely by sEMG data.

Company name	Core products	Underlying technology	Key Feature	Recent progress
BrainCo 強腦科技	-Bionic hands. -Bionic legs. -Sleep induction devices. -Cognitive training systems.	-Non-invasive electroencephalogram sensing. -Surface electromyography arrays. -Deep learning decoding. -Dual-beat sound modulation.	The bionic hand opens in 0.8 seconds. Sensors capture muscle signals with low latency. Devices integrate heart rate and blood oxygen monitoring. Systems output mindfulness indices directly.	The firm mass-produced consumer and medical hardware. Products are sold through major online platforms. The bionic hand assists users in typing and playing instruments. The company raised 2 billion RMB in January 2026.
Fourier Intelligence 傅利叶智能	-Galileo biomechanical system. -GRx series humanoid robots. -Rehabilitation exoskeletons.	Multimodal systems capture active movement intentions using wireless sEMG electrodes. Control algorithms compute integrated EMG and difference EMG to quantify energy trends. Systems provide assist-as-needed force feedback based on these signals.	The platform integrates 19 distinct sensor nodes. It coordinates force plates and motion capture with muscle signals. It provides adaptive impedance control for active patient interaction.	The company finalised an 800 million RMB E-series financing round in January 2026. The firm deployed complex environmental simulation platforms. They integrated embodied AI into active rehabilitation.
OYMotion 傲意科技	-Bionic hands. -Neural rehabilitation exoskeletons. -Humanoid robot hands. -Electroencephalographs.	-Brain-muscle-machine interfaces. -Signal amplification structures. -Multi-channel filtering matrices. -Artificial intelligence pattern recognition.	Flexible electrode arrays reduce signal distortion. The system processes complex electromyography data efficiently. Medical motor technologies are adapted for robotic parts. The company manufactures its own core microchips.	The firm constructed a manufacturing factory in Zhuhai. Robot hand component shipments increased fivefold. Overseas commercial sales grew faster than domestic operations. The company secured 150 million RMB in March 2026.

For literatures and conferences

Common Consensus on Problems:

- Signal Degradation: Model accuracy drops significantly across different days and users.
- Physical Interference: Skin temperature changes, and small electrode shifts will affect static classification models.
- Placement limits: The wrist provides unstable signals. The upper forearm is the required standard for sensor placement.

Different solutions:

- Combining both physical model/rules and Deep learning models.
- High-Density & Deep Learning: Uses 64+ channels and complex networks, requires high computational capacity.
- Using simple time-domain features (RMS, MAV, WL) and lightweight classifiers (SVM, LDA).
- Cross-modal supervision uses optical cameras (vision oracles)

Similarities:

- The basic theoretical derivations in the literature mostly rely on commercial high-fidelity wet electrodes or medical-grade array devices; while any project involving human-computer interaction and real-time 3D virtual demonstrations tacitly compromises on the portability of dry electrode armbands, such as the discontinued Myo.
- In response to the high impedance and triboelectric artifacts of dry electrodes, all system engineering incorporates bandpass and notch filters (cutting off 50Hz) and uses time-domain features as the main input to the classifier, thus eliminating dependence on high-frequency signals.

For GitHub open source

Similarities:

- All three repositories strictly divide the control system into independent stages: data acquisition, signal processing, and model prediction.
- Python is the standard for backend development across all projects.

Differences:

- Deep Learning & Spatial Grids (Facebook, Tsagkas): These systems use PyTorch or Keras to run CNNs. They rely on high spatial resolution (multi-channel armbands or HD-sEMG arrays) to automatically extract patterns from raw data.
- Explicit Feature Engineering (LibEMG): This system uses traditional Machine Learning. It manually extracts mathematical features (like RMS or MAV) from the signal envelope before classification. It is designed to be lightweight.

For the three companies that received the most funding in this sector over the past year:

Similarities:

- Both BrainCo and OYMotion use dry electrode eight-channel electromyography armbands for sEMG control. OYMotion, in particular, started with electromyography sensor technology, and its related products cover a range of products including wet electrodes.
- In addition to the sEMG sensor, all electromyography armbands are equipped with 9-axis Inertial Measurement Units.
- All three companies' commercial products have abandoned reliance on external high-performance computers. They all require feature extraction and pattern recognition algorithms (such as lightweight machine learning models) to be directly burned into and deployed in the microcontroller of the device (embedded system) to achieve extremely low interaction latency.

Differences:

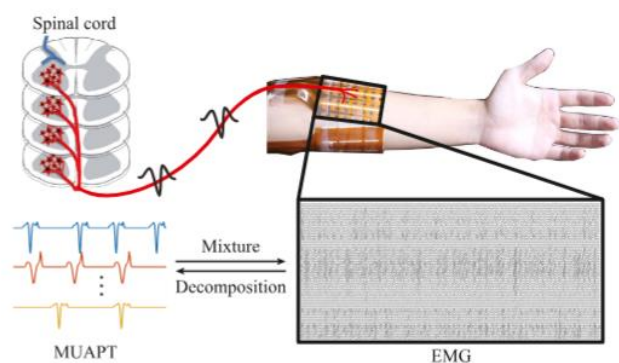
- BrainCo and OYMotion's core electromyography products are Bionic Hands, their systems decode sEMG signals into discrete gestures or positions to drive bionic hands for amputees. Meanwhile, the Fourier Focuses on Dynamic Impedance Control. Their exoskeletons use sEMG to estimate muscle force and energy trends, providing dynamic force feedback for clinical rehabilitation.
- Fourier Intelligence achieves true physical sensor fusion, combining sEMG with force plates and motion capture to validate complex biomechanics. These supports using an external physical oracle (like vision) to calibrate sEMG. BrainCo possesses both EEG and sEMG technologies but deploys them as separate commercial product lines rather than a unified, real-time sensor fusion system.

Project Expectations:

For the dual-channel low fidelity hardware baseline of this project, in the action dimension, limited by the physical resolution of the dual channels, 4 macroscopic gestures are selected to map the actions: Hand open, Grip, Pitch, wrist extension (state change).

The system will mainly extract lightweight temporal features such as RMS and MAV based on the literatures and conferences, to ensure the real-time response of the classification algorithm on the embedded device.

In the subject dimension, current hardware inherently lacks cross-subject generalisation ability. therefore, the goal will include one completely new, untrained subject as a control experiment.



(A)



(B)



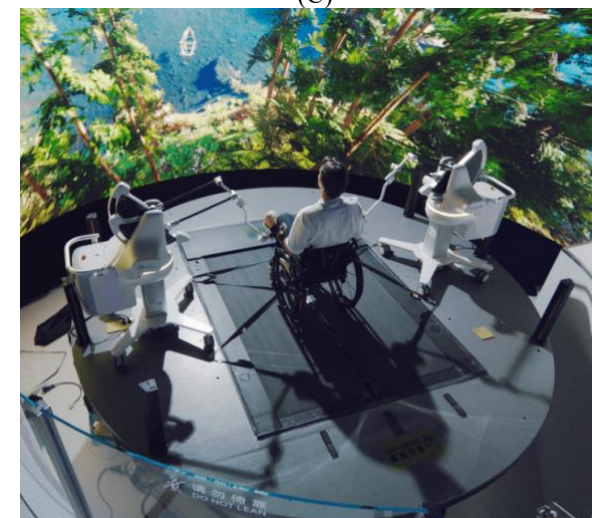
(C)



(D)



(E)



(F)

Fig. 1. (A) High-Density-sEMG arrays. (B) BrainCo EMG armband. (C) OYMotion gForce Ultra armband. (D) OYMotion gForcePro+ armband. (E) OYMotion gForceOct Eight-channel wireless electromyography device. (F) Fourier MetaMotus™ Galileo Biomechanics Analysis and Rehabilitation Platform.

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